aivancity

La Grande Ecole de l'IA & de la Data

Paris Île-de-France · Nice Côte d'Azur

PGE 2

Machine Learning

Session 1 · Data pipelines and core principles

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Today's objectives

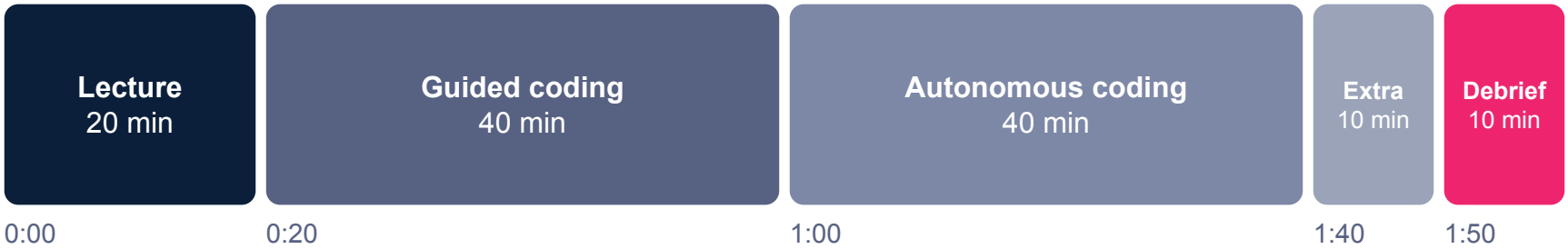
- Name the parts of a supervised model: X, y, w, prediction
- Build a clean, leakage-free data pipeline on real data
- Split data into train, validation and test sets correctly
- Compare any model to a baseline



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Session 1

How today works



Rule of the course: short lecture, long coding. About 80% of the time you write code.

Notebook: S1_data_pipeline.ipynb on Blackboard. Core part for everyone, extension part if you finish early.

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Part 1

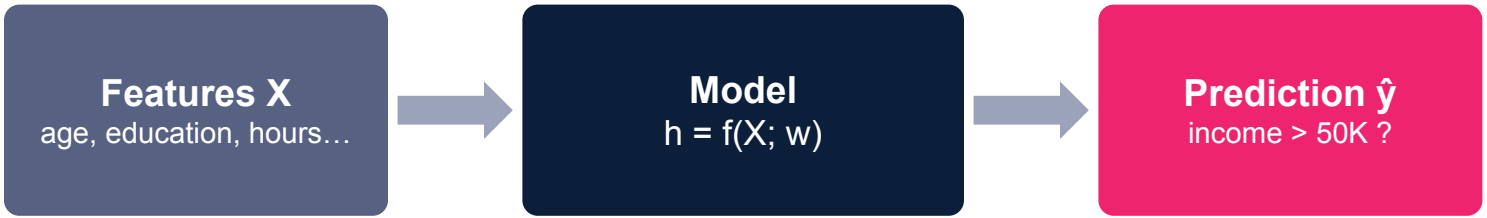
Lecture: from raw data to a model

20 minutes · 9 slides



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A supervised model learns a function from examples



X
features: what we know

y
target: what we want to predict

w
parameters: what the model learns

ŷ = h(X)
prediction, compared with y by a loss

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Session 1

Two families of tasks this semester

Regression: y is a number

Price of a flat · rentals per hour · energy output

Sessions 2 and 3

Classification: y is a category

Spam or not · income > 50K · machine failure

Sessions 4 to 7

Supervised

We have examples with the answer y. This is the whole of semester 1.

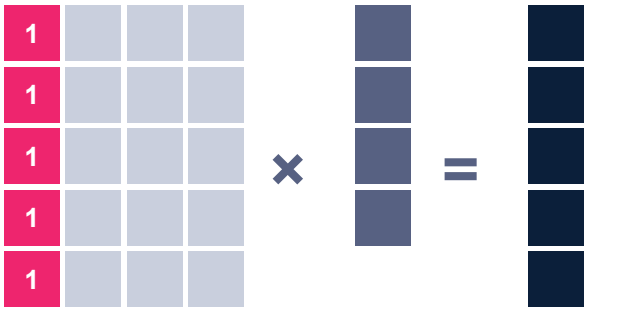
Unsupervised

No y: we look for structure (clusters, segments). Semester 2.

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Always know the shape of your arrays



X (m, n+1) **w** (n+1,) **h = Xw** (m,)

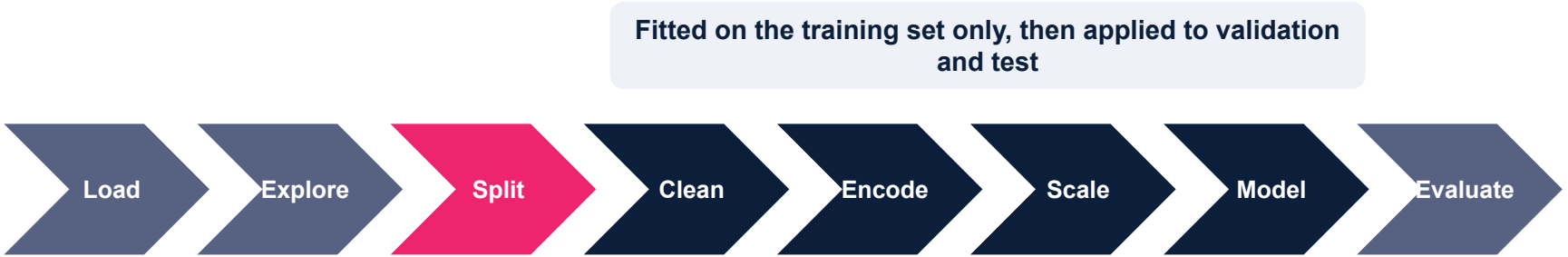
- **m** rows = samples
- **n** columns = features
- **Column of ones** = the bias: w_0 is learned like any other weight
- **y** has shape (m,), not (m, 1): otherwise Xw − y silently becomes (m, m)

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Session 1

The data pipeline: split before you learn anything



- **Explore** the full data to understand it, but never compute a statistic you will reuse (mean, mode, min, max) before the split.
- **Leakage** = information from validation or test data reaching training. Results look great, then fail in production.

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Session 1

Cleaning: four problems to look for

Missing values

NaN, empty strings, or '?' in raw files. Drop, impute (mode, median) or add an 'Unknown' category.

Duplicates

Identical rows inflate some patterns and can leak between train and test.

Outliers

Extreme but real values (capital-gain) vs data errors. Decide, then justify.

Sentinel values

A number meaning 'unknown' or 'capped': 99999, -1, -200. Treat as missing.

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Encoding and scaling

One-hot encoding

workclass →	Private	Self-emp	Gov
Private	1	0	0
Gov	0	0	1
Self-emp	0	1	0

- A category seen only in test? Use OneHotEncoder(handle_unknown="ignore").
- Never encode categories as 1, 2, 3: it invents an order.

Scaling numeric features

$$z = \frac{x - \mu_{train}}{\sigma_{train}}$$

Standardisation: mean 0, std 1 (default choice)

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Min-max: values in [0, 1]. Statistics always come from train.

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Session 1

Train, validation, test: who sees what



fit the model and the preparation

choose settings, compare models

final score, used once

```
X_train, X_tmp, y_train, y_tmp = train_test_split(X, y, test_size=0.4,
                                                  stratify=y, random_state=42)
X_val, X_test, y_val, y_test = train_test_split(X_tmp, y_tmp, test_size=0.5,
                                                stratify=y_tmp, random_state=42)
```

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scikit-learn does the bookkeeping for you

```
num = ["age", "education-num", "capital-gain",
       "capital-loss", "hours-per-week"]
cat = ["workclass", "marital-status", "occupation",
       "relationship", "race", "sex", "native-country"]

cat_pipe = Pipeline([
    ("impute", SimpleImputer(strategy="most_frequent")),
    ("onehot", OneHotEncoder(handle_unknown="ignore"))])

prep = ColumnTransformer([("num", StandardScaler(), num),
                          ("cat", cat_pipe, cat)])

prep.fit(X_train)                # learn on train only
X_val_ready = prep.transform(X_val)
```

- **fit** learns the statistics (means, modes, categories)
- **transform** applies them to any set
- Today you write it by hand first, then rebuild it with these tools and check both match.

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Session 1

Always start with a baseline

76 %

accuracy of a 'model' that always predicts ≤ 50K on Adult income

A baseline is the dumbest reasonable model:

- regression: predict the mean of y_train
- classification: predict the most frequent class
- If your model does not beat it, it has learned nothing useful.
- Is 76 % accuracy good? We answer properly in Session 5.

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Part 2

Your turn: build the pipeline

80 minutes of coding · S1_data_pipeline.ipynb



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Session 1

Today's dataset: Adult income (1994 US census)

48,842
people (rows)

14
features

24 %
earn more than 50K

3
columns with missing values

```
from sklearn.datasets import fetch_openml
df = fetch_openml("adult", version=2,
                  as_frame=True).frame
```

Source: UCI Machine Learning Repository, CC BY 4.0

Watch out

- fnlwgt is a survey weight, not a feature
- education duplicates education-num
- race and sex are sensitive attributes: 5-minute discussion

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Notebook roadmap

1 · Load and inspect
2 · Explore: distributions, target balance, capital-gain
3 · Clean: missing values, duplicates, columns to drop
4 · Split 60 / 20 / 20 with stratify
5 · Prepare by hand: fit_prepare / apply_prepare
6 · Rebuild with Pipeline + ColumnTransformer
7 · Baseline with DummyClassifier

Every function has assert checks: green means you can move on.

Session 1

Debrief: three quick checks (show of hands)

1. Predicting tomorrow's bike rentals: supervised or unsupervised? Regression or classification?

2. You compute the imputation mean on the full dataset before splitting. What is the problem?

3. X has 1,000 rows, 12 features and a column of ones. What is the shape of w?

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Questions?

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